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GitHub Link	https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO2

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning
Faculty : Dr. Arul Raj A.M

COURSE CODE: MLA0405
Slot : D

COURSE OUTCOME COVERED

CO 2:Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

Assessment Tool Weightage: 40%

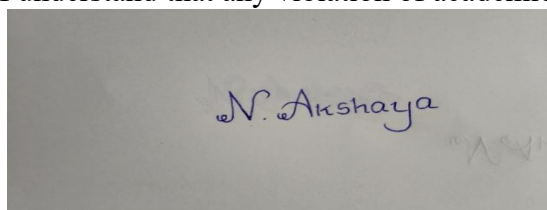
Duration: 90 minutes

Marks: 25

Code of Conduct:

I **N.AKSHAYA(192425129)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: N.AKSHAYA

Scenario-Based Questions CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Answer:

Recommended Technique: Autoencoder (Unsupervised Learning)

Justification:

An Autoencoder is an unsupervised neural network designed to learn the normal patterns present in unlabeled data. Since the hospital has thousands of ECG records without labels, an Autoencoder is an ideal choice. It compresses ECG signals into a lower-dimensional representation using an encoder and reconstructs them using a decoder. Normal ECG signals are reconstructed with low error, whereas abnormal ECG signals produce high reconstruction error, enabling anomaly detection.

Workflow:

1. Collect and preprocess unlabeled ECG records.
2. Normalize ECG signals to improve learning.
3. Train the Autoencoder using normal ECG patterns.
4. The encoder extracts important features into a latent representation.
5. The decoder reconstructs the original ECG signal.
6. Compute the reconstruction error.
7. If the error is greater than a predefined threshold, classify the ECG as abnormal; otherwise, classify it as normal.

Advantages:

- Works without labeled datasets.
- Learns hidden ECG features automatically.
- Detects abnormal heart patterns accurately.
- Reduces manual effort for medical professionals.
- Supports early diagnosis of heart diseases.

Disadvantages:

- Requires a large amount of training data.
- High reconstruction error threshold selection can be difficult.
- May fail to detect very subtle abnormalities.
- Training can be computationally expensive.
- Performance depends on data quality and preprocessing.

Conclusion:

The Autoencoder is the most suitable technique for detecting abnormal ECG patterns because it learns normal ECG features from unlabeled data and identifies anomalies using reconstruction error. This approach enables accurate, efficient, and automated detection of abnormal heart conditions without requiring labeled datasets.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Answer:

An Autoencoder is an unsupervised neural network that learns a compressed representation of high-dimensional customer purchase data while preserving important information.

Working:

1. Collect customer purchase records.
2. Normalize and preprocess the data.
3. Feed the data into the Autoencoder.
4. The encoder compresses the data into latent features.
5. The decoder reconstructs the original purchase data.
6. Train the model by minimizing reconstruction error.
7. Use the compressed features as input for customer segmentation algorithms such as K-Means.

Advantages:

- Reduces dimensionality.
- Removes redundant information.
- Improves clustering accuracy.
- Reduces computational cost.
- Helps identify meaningful customer groups.

Disadvantages:

- Important information may be lost during compression.
- Requires careful tuning of network architecture.
- Computationally intensive for large datasets.
- May not perform well with noisy or incomplete data.
- Reconstruction quality affects feature representation.

Conclusion:

Autoencoders effectively reduce the dimensionality of customer purchase data by extracting meaningful features while preserving important information. The compressed data improves customer segmentation accuracy, reduces computational cost, and enhances business decision-making.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Answer:

A shallow neural network cannot effectively learn complex facial features. Increasing network depth and using suitable activation functions improves recognition accuracy.

Recommended Changes:

1. Increase the number of hidden layers.
2. Increase the number of neurons in each hidden layer.
3. Use ReLU activation in hidden layers.
4. Use Softmax activation in the output layer.
5. Apply Batch Normalization.
6. Use Dropout to reduce overfitting.

Benefits:

- Learns complex facial features.
- Improves recognition accuracy.
- Faster model convergence.
- Better generalization.
- Handles variations in lighting and facial expressions.

Conclusion:

Increasing the depth of the neural network and using appropriate activation functions such as ReLU significantly improves facial recognition performance. These architectural changes enable the model to learn complex facial features, resulting in higher accuracy and better generalization.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer:

Restricted Boltzmann Machines (RBMs) are unsupervised neural networks that learn hidden relationships between users and movies.

Workflow:

1. Create a user-movie rating matrix.
2. Train the RBM without labeled data.
3. The visible layer stores movie ratings.
4. The hidden layer learns user preferences.
5. Extract latent features.
6. Predict ratings for unseen movies.
7. Recommend movies with the highest predicted ratings.

Advantages:

- No labeled data required.
- Learns hidden user preferences.
- Handles sparse datasets efficiently.
- Provides personalized recommendations.
- Improves recommendation accuracy.

Disadvantages:

- Slow training compared to modern deep learning models.
- Difficult to optimize for very large datasets.
- Sensitive to parameter initialization.
- May converge slowly or get stuck in local minima.
- Less effective than newer recommendation techniques such as deep neural networks.

Conclusion:

Restricted Boltzmann Machines provide an effective unsupervised approach for recommendation systems by learning hidden user preferences from viewing history. This allows the streaming service to deliver personalized and accurate movie recommendations without requiring labeled data.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Answer:**Proposed Solution**

A deep learning model combining **representation learning (Autoencoder)** and a **Deep Neural Network (DNN)** can automatically detect defective products from sensor data.

Workflow:

1. Collect real-time sensor readings.
2. Preprocess and normalize the data.
3. Train an Autoencoder to learn normal sensor patterns.
4. Extract latent features using the encoder.
5. Feed the latent features into a Deep Neural Network.
6. Classify products as normal or defective.
7. Generate alerts for defective products.

Advantages:

- Detects defects in real time.
- Improves product quality.
- Reduces manual inspection.
- Minimizes production losses.
- Supports industrial automation.

Disadvantages:

- Requires large amounts of sensor data for training.
- High computational and hardware requirements.
- Performance decreases with poor-quality or noisy sensor data.
- Model maintenance and retraining are needed as production conditions change.
- Initial implementation cost can be high.

Conclusion:

A deep learning solution using representation learning and neural networks enables accurate real-time defect detection from sensor data. This approach improves product quality, reduces manual inspection, minimizes production losses, and supports efficient industrial automation.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand